

SRViT: A Robust Online Encrypted Traffic Classification Based on Vision Transformer

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Abstract—The dramatic rise in encrypted traffic brings huge challenges to traditional traffic classification methods. Deep learning-based traffic classification methods have been demonstrated to significantly improve performance. However, the following limitations remain: i) It is challenging to concurrently focus on both global and local information in traffic flows, resulting in the absence of important information. ii) The existing methods relying on temporal information suffer from low robustness in case of packet disordering or loss. iii) The use of multi-layer encryption and random routing in Tor technology poses more challenges for traffic identification. In this paper, we propose a novel ViT-based model for more accurate encrypted traffic classification, called SRViT to overcome the above challenges. Firstly, SRViT proposes a novel mechanism of multi-size patch division to learn comprehensive hidden knowledge and dependencies between packets. Secondly, we propose a self-attention operation with a relative position bias to learn the relative position relationship. After that, an incremental update mechanism is proposed to adapt to dynamic changes in the real traffic environment. At last, the comprehensive experiments on 5 real-world encrypted traffic datasets are carried out. The experimental results indicate that SRViT outperforms the state-of-the-art methods with an average accuracy improvement of 24.62% while keeping higher robustness and execution efficiency.

Index Terms—Encrypted traffic classification, vision transformer, Tor traffic.

I. INTRODUCTION

NETWORK traffic classification is an important task of applications or Web services identifying in network

management and cyberspace security. In network management, traffic priority classification can help ensure the differentiated Service-Level Agreement (SLA) requirements [1], [2], [3], [4]. In cyberspace security, accurate traffic identification can prevent the spreading of a significant variety of malicious attacks [5], [6], [7], [8], [9]. Nowadays, with the widespread adoption of encryption for privacy protection, encrypted traffic has risen dramatically and occupies a large share of network traffic [10], [11], [12]. Therefore, encrypted flow classification has become a topical issue and has attracted extensive attention from both industry and academia. How to capture the robust patterns implied in diverse encrypted traffic and support accurate and generalized traffic classification is crucial for achieving high network security and effective network management.

However, malware traffic and cybercriminals can consequently evade monitoring systems through privacy-enhancing encryption techniques such as Tor, VPN, etc., which lead to the ineffectiveness of traditional rule/DPI-based classification approaches due to the randomized communication contents [13]. Some studies [14], [15], [16], [17], [18] combine machine learning algorithms with statistical features manually extracted from raw traffic to handle encrypted traffic. And, Lichy et al. [19] demonstrated that, in some cases, classical machine learning models (e.g., random forests) perform better in malware traffic classification tasks. However, these methods are highly dependent on expert-designed features and have limited generalization capabilities. Recently, deep learning methods automatically learn complex patterns from raw traffic and achieve significant performance improvements. Some methods [20], [21], [22], [23] extract information, such as packet size or packet intervals sequence, from a temporal perspective. However, these approaches may suffer from a drastic drop in performance under packet disorder and packet loss. Some researchers [24], [25], [26] take the encrypted traffic classification as a computer vision task through transforming encrypted traffic flows to images. These methods focus on local information extraction of each traffic flow while ignoring the global and contextual information understanding. However, fixed convolution kernels can't adapt to dynamic changes in encrypted traffic patterns.

Unlike CNNs, a Transformer [27] based on the self-attention mechanism can take into account both local interactions and long-distance dependencies through the dynamic computation of attention scores with appropriate generalization biases according to different task goals. ET-BERT [28] introduces a Transformer-based large model into encrypted traffic

Received 4 September 2024; revised 29 May 2025 and 22 September 2025; accepted 28 May 2026; approved by IEEE TRANSACTIONS ON NETWORKING Editor D. Pei. Date of publication 3 June 2026; date of current version 9 June 2026. This work was supported in part by the National Major Science and Technology Program of China under Grant 2026ZD1307700; in part by the Major Scientific Instruments and Equipment Development Project of the National Natural Science Foundation of China under Grant 62427809; in part by the National Natural Science Foundation of China under Grant 62572187, Grant 62472168, Grant 62402234, Grant 62202237, and Grant NY223168; in part by the Key Program of Natural Science Foundation of Jiangsu Province under Grant BK20243053; in part by the Research Foundation of Education Bureau of Hunan Province, China, under Grant 25B0452; in part by the Open Project of State Key Laboratory of Internet Architecture under Grant HLW2025MS24; and in part by the State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, under Grant CLQ202506. (Corresponding author: Fu Xiao.)

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Digital Object Identifier 10.1109/TON.2026.3699540

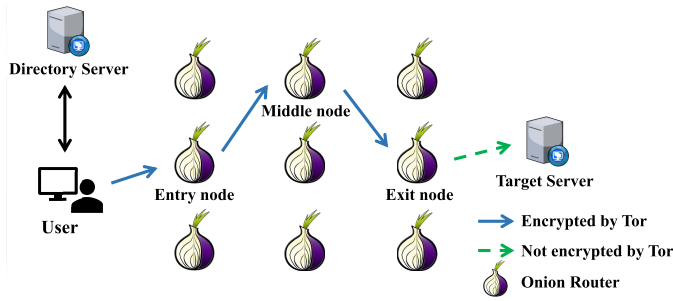


Fig. 1. Tor system structure.

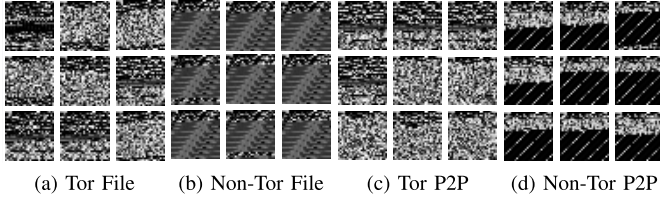


Fig. 2. The visualization results of Tor-encrypted traffic compared to non-Tor-encrypted traffic.

classification tasks for performance improvement. NetGPT [29] proposes a generative pre-trained transformer for network traffic. YaTC [30] introduces a masked autoencoder (MAE) based traffic transformer with multi-level flow representation for traffic classification. However, most of these methods inevitably incur high computational complexity in the flow representations learning, pre-training, and fine-tuning process.

With increasing traffic arrival rates, excessive computational complexity would impede the deployment of traffic identification methods. In this paper, we aim to propose a transformer-based traffic classification method to utilize both temporal and 2-dimensional spatial information while requiring substantially fewer computational resources. ViT (Vision Transformer) [31] is a natural choice. It applies a pure transformer directly to sequences of image patches and attains excellent classifying results [32].

However, there exist some compatibility problems when directly introducing ViT into traffic classification tasks, especially for Tor traffic identification [7]. The Tor [33] system's architecture is shown in Fig. 1. The user randomly selects three onion routes and establishes a circuit with these nodes. Multiplexing multiple circuits per connection. The client encrypts the packets in three layers and decrypts them sequentially by each relay node. The path randomization and multi-layer encryption mechanisms may result in different numbers of encryption layers in received packets of varying traffic flows or applications for an onion router. Fig. 2 shows the visualization results of Tor-encrypted traffic compared to non-Tor-encrypted traffic. The figure clearly shows that there is an obvious difference in the visualization of traffic between the different types of traffic. However, the difference between traffic encrypted with Tor seems to be much weaker than the traffic without using Tor encryption. These characteristics of multi-layer encryption, highly obfuscated payloads, random routing, and traffic pattern uncertainty impair the effectiveness of fixed patch-wise interaction learning in ViT.

To address the above challenges, we propose SRViT (multi-size relative ViT), a novel ViT-based model for more accurate

encrypted traffic classification. To be specific, we introduce a multi-size patch overlapping division mechanism to enable the model to more comprehensively learn the characteristics of encrypted traffic. In addition, we add relative position bias to the self-attention mechanism in the encoding layer to learn the relative position relationship of encrypted traffic. Finally, we also incorporate an incremental update mechanism to enable the model to adapt the dynamically changing traffic quickly. Compared with methods, our paper makes the following contributions:

- We introduce ViT into encrypted traffic classification tasks and propose a novel end-to-end identification model, which transforms traffic flows into image patches and dynamically learns the local/global dependencies between packets.
- We propose a novel multi-size patch mechanism to further comprehensively learn the hidden features and resolve the multi-layer encryption, highly obfuscated payloads, random routing, and traffic pattern uncertainty problems. In addition, we propose a self-attention operation with a novel relative position bias, which can dynamically adjust the attention scores and learn the relative position relationship.
- We design a novel incremental update to adapt to the dynamically changed traffic by only updating a small proportion of parameters when a sharp drop in identification accuracy occurs.
- We conduct a series of experiments over 5 real-world encrypted traffic datasets. Compared with the state-of-the-art methods, SRViT achieves the highest classification accuracy, with an average improvement of 25.04% in accuracy, while keeping higher robustness and execution efficiency.

The rest of the paper is organized as follows. We first summarize the related work of traffic classification in Section II and introduce the preliminary work in Section III. We then present the technical details of our novel SRViT framework in Section IV. After that, we evaluate the performance of our proposed model by conducting experiments on real-world datasets in Section V. Finally, we present a detailed discussion of SRViT in Section VI and conclude this paper in Section VII.

II. RELATED WORK

In this section, we briefly introduce these related works and classify them according to conventional traffic classification, machine learning-based methods, and deep learning-based methods as follows.

A. Conventional Traffic Classification

Traffic classification primarily involves port-based methods and deep packet inspection (DPI)-based methods. The port-based approach [34] is used to identify the application type by a given list of ports provided by the Internet Assigned Numbers Authority (IANA). However, this method fails in cases of dynamic port allocation and the use of generic communication protocol ports [35]. The port allocation and

port masquerading make it no longer applicable in traffic classification.

The payload-based approaches exploit the specific signature in the payload for matching. Roughan et al. [36] used statistical signatures to classify P2P application traffic. Meanwhile, Keralapura et al. [37] provided a self-learning traffic classifier for identifying P2P traffic in high-speed networks with application payload signatures.

However, due to the emergence of traffic encryption and encapsulation, and the concern of privacy protection in recent years, most payload methods become inapplicable anymore. Thus the advent of encrypted traffic makes network traffic classification more challenging.

B. Encrypted Traffic Classification With Machine Learning

Different from the packet inspection method under plain text traffic, which fails when the traffic is encrypted, some research suggests using unencrypted protocol field information. For example, FlowPrint [15] extracts the device, certificate, size, and temporal features from the header field of each packet to make the representation of each flow and then constructs a fingerprint database. Finally, through the techniques of clustering and cross-correlating, the identification of each type of flow is achieved. However, these fingerprints are more likely susceptible to the noise from the network environment, leading to the loss of useful knowledge especially in virtual communication networks.

With the advent of machine learning techniques, recent research has employed machine learning techniques to deal with encrypted traffic classification. Researchers have mainly worked on the extraction of effective and stable information from the packet header. Namely, considering how to construct enough valid features rather than extracting signatures from packet payload. There are two types of features commonly used in encrypted traffic classification: statistical features and sequential features.

Statistical Features: Combining with various traditional machine learning algorithms, most encrypted traffic classification problems are addressed using statistical features. Examples include logistic regression, random forest, and support vector machine. For instance, AppScanner [38] extracts statistical features based on packet length and uses a random forest classifier. Through the analysis of encrypted network traffic, it proposes a robust method for identifying smartphone applications with the concept of burst and traffic statistics features. STM-DT [18] is a method for classifying network attacking traffic based on the decision tree. HyperVision [6] is a machine learning-based method for detecting unknown patterns in encrypted malicious traffic using an in-memory graph. NetBeacon [39] deploys machine learning models in the Intelligent Data Plane (IDP) for malicious traffic detection. However, these methods often rely on much expertise and specialized knowledge, resulting in the difficulty in generalizing to the flows from constantly emerging applications and the traffic in complex network environments such as packet disorder and packet loss.

Sequential Features: The work [40] proposes to use the Markov transformation matrices to learn the temporal

knowledge from traffic sequences for the first time. Specifically, it uses the sequence of message type to construct a first-order Markov model with MGP (maximum generation probability). Building on this work, Shen et al. improve the classification performance under the second-order Markov model [41], [42]. In addition, FU et al. [43] extract packet length and time delay sequences to construct a Hidden Markov Model. Due to inefficiency in capturing long-term relationships for lower-order Markov models and high computing complexity for higher models, these methods are likely to be unsuitable for complex networks such as those with high throughput. These methods are incapable of capturing long-term relationships and at the same time segmenting the feature learning and classification process.

C. Encrypted Traffic Classification Based on Deep Learning

Deep learning has been well used in the fields of image processing and natural language processing, and researchers have made multiple attempts to apply them in encrypted traffic. The use of supervised deep learning for encrypted traffic classification has become a popular approach. Beauty [44] and DF [24] are based on convolutional neural networks (CNNs). An autonomous model updating framework [26] is proposed to perform accurate traffic classification using improved CNN classifiers. FS-Net [20] uses recurrent neural networks (RNNs) to automatically extract representations from the raw data packets of encrypted traffic. Deeppacket [25] employs stacked autoencoder and 1D convolutional neural networks to achieve end-to-end traffic classification. GraphDApp [45] introduces Traffic Interaction Graphs (TIGs) to represent each individual encrypted flow, serving as a method for DApp fingerprinting using graph neural networks (GNNs). ET-BERT [28] proposes a pre-training framework where the downstream task uses a transformer model for encrypted traffic classification. In addition, the work [46] proposes Mini-FlowPic, using augmentations for few sample traffic classification, which utilizes CNN and comparative learning techniques to achieve efficient and accurate cryptographic traffic classification with limited labeled data. EC-GCN [47] classifies encrypted traffic flows based on multi-scale graph convolutional neural networks. DMSTG [48] employs a dynamic learning module to timely estimate the graph Laplacian for GCN and a self-attention module to adaptively weight traffic data, enabling the effective fusion of multi-view spatio-temporal features to capture dynamic changes in traffic patterns. RBLJAN [49] proposes Robust Byte-Labeled Joint Attention Network, which is an efficient and robust deep learning-based framework for both packet-level and flow-level encrypted network traffic classification. RBLJAN shows superior performance in terms of detection speed and robustness.

Our approach mainly differs from existing ones in the following aspects: Firstly, we introduce ViT into encrypted traffic classification tasks and an overlapping multi-size patch division mechanism that allows the model to learn the features of encrypted traffic more comprehensively and improves the efficiency of Tor-encrypted traffic classification. In addition, we add relative position bias to the self-attention mechanism in the encoding layer to adjust the self-attention score and

learn the relative position relationship of encrypted traffic dynamically to enhance the robustness of the model. Finally, we incorporate an incremental update mechanism that updates only a small number of parameters when the model's recognition accuracy suffers a large degradation, allowing the model to quickly adapt to dynamically changing traffic reaching online classification.

III. PRELIMINARIES

A. Tor and Traffic Analysis

Tor [33], also known as The Onion Router, is an open-source network designed to facilitate anonymous communication over the Internet. Its primary goal is to obscure the user's real IP address and the content of their communication by routing and encrypting the user's data over the Internet. Tor system consists of users, onion proxies, onion routers, directory servers, and target application servers. The architecture is shown in Fig. 1.

The Tor network includes thousands of relay nodes (onion router, OR) operated by volunteers worldwide, using a multi-hop proxy mechanism for user communication privacy. The client runs an Onion Proxy to randomly select onion routers from the directory server and establish a path that we call a circuit. Typically, this circuit consists of three ORs: Entry Node (or Guard Node), Middle Node, and Exit Node, and then the user establishes a circuit with these nodes hop by hop. During data transmission, OP encrypts user data in multiple layers and decrypts it sequentially by each relay node. Onion router uses public keys to establish TCP connections secured by TLS, multiplexing multiple circuits per connection. When a client accesses different target servers, a new circuit is built in real-time. In addition, the client OP re-selects a new circuit every 10 minutes. Therefore, the same node may appear in different locations on multiple circuits simultaneously and the packets through it may vary in the number of encrypted layers. Through providing sender privacy for internet services to its users, Tor has become a popular and low-latency anonymity network. The characteristics in the Tor network can be summarized into the following three aspects:

- **Multi-layer Encryption (Onion-like encryption):** Each OR decrypts only one layer of encryption corresponding to it to reveal the next relay node in the path and forwards the remaining encrypted data to it.
- **Fixed-size Cells:** Tor divides data into fixed-sized onion-encrypted units called cells (typically 512 bytes) before transmission. This standardization helps to obfuscate the actual size and structure of the data, making it harder to analyze and correlate by size alone.
- **Random Routing:** The circuit through which data travels is randomly selected for each session, and it changes periodically. This dynamic and unpredictable routing makes it difficult for adversaries to track users' activities or correlate traffic between relays.

Although encryption makes the payload look cluttered and unsemantic, Sengupta et al. [50] exploit the randomness difference between different ciphertexts to distinguish different applications, which suggests that the encrypted traffic

is not perfectly random and implicit patterns exist. Therefore, deep learning methods can be used to extract the implicit deep information in raw traffic. Even in encrypted and anonymized environments like Tor, deep learning models, especially those designed for sequence analysis like Transformer, can learn to extract and identify correlations in these subtle packet sequences that are not immediately apparent to human analysts, and identify traffic when traditional methods fail.

B. Vision Transformer

Transformer based on self-attention mechanism has made a great success in the field of NLP. Although the Transformer structure has become a standard in the NLP, its application in the computer vision field is still very limited. ViT (vision transformer) is a model that directly applies the Transformer to image classification and achieves good performance.

Transformer consists of multi-head self-attention layers and feed-forward neural network layers. The self-attention layer is used to encode the sequence and the feed-forward neural network layer is used to process the features at each position in the sequence. The self-attention mechanism calculates the attention scores between each pair of tokens in the input data, thus enabling the model to capture complex dependencies regardless of their distance in the sequence. Transformer uses the self-attention mechanism for greater parallelism and computational efficiency when dealing with long sequences and large-scale data. Multi-head attention splits the input into multiple subsets and computes the attention scores for each subset independently, which allows the model to learn information in multiple subspaces and enhances the model's sensitivity to different locations, thus enabling the model to capture different types of relationships and patterns in the data simultaneously.

ViT divides the input image into a sequence of non-overlapping patches, and then the sequence of patches is treated as tokens, which can be fed into the original Transformer to accomplish the classification task. Since the latent feature extraction of encrypted traffic packets is more suitable as a CV task [30]. ViT can utilize both temporal and 2-dimensional spatial information, the model is simple, effective and highly scalable. Therefore, this paper makes ViT the backbone network but devises a dedicated ViT structure to make it more suitable for Tor-encrypted traffic classification. First, we design an overlapping multi-size patch mechanism to learn the features of Tor encrypted traffic flows more comprehensively. Second, we propose a relative position bias in the multi-headed self-attention mechanism to dynamically adjust the attention score and learn the relative position relationship in each encrypted traffic flow. As detailed in Section IV.

IV. PROBLEM DESCRIPTION AND MODEL

In this section, we introduce the encrypted traffic classification problem and the basic framework of our proposed model. After that, we describe our flow representation and classifier structure in detail.

TABLE I
LIST OF NOTATIONS

Notation	Meaning
n	The number of first n bytes intercepted by the model
x	The input image
p	The number of patches
Z	The number of categories
Y	The predicted label by SRViT
N	The number of patches generated by the first layer
M	The number of patches generated by the second layer
P	The patch size
S	The stride length
d	The embedding dimension
d_{model}	The vector dimension in positional encoding
H	The height of the image
W	The width of the image
C	The number of channels of the image
L	The number of layers stacked with RPB self-attention and MLP block
h	The number of heads of the RPB self-attention
X_l	Output of layer l of the self-attention
PE	Calculated relative position

A. Problem Definition

Our paper focuses on building an end-to-end encrypted traffic identification model to accurately categorize malicious encrypted traffic by taking each traffic flow as input. To maintain consistency in input length, we only select the first n bytes of each flow and formulate it as $(x_1, x_2, \dots, x_p)$ after patch splitting, where p is the number of patches and x_i denotes the i th patch token. The output is a vector $Y \in \mathbb{R}^Z$, where Z denotes the number of categories. If a traffic flow belongs to the j th class, then $Y_j = 1$, otherwise $Y_j = 0$. The notations used in this paper are shown in Table I.

B. Overview of The Model

Encryption typically makes traffic appear more random and less predictable. Multi-layer encryption aggravates this effect, with each encryption layer changing the statistical properties of the traffic, and patterns recognizable in single-layer encrypted traffic are further obscured in multi-layer encrypted traffic. Because of the characteristics of multi-layer encryption, highly obfuscated payloads, and random routing, the similarity between the same category of Tor traffic flows is nearly weak and the difference between different categories is not obvious, which makes traditional encrypted traffic identification algorithms have not performed optimally in Tor encrypted traffic identification.

Furthermore, with the explosive growth of massive Internet traffic, the number of packets required to be analyzed is constantly increasing. As a result, the performance of traditional identification algorithms on large scale training samples and features is far from satisfactory. Although the model based on Transformer has achieved great improvement in Tor traffic identification accuracy, its high computational complexity incurred during traffic representation learning, pre-training, and fine-tuning impedes the practical deployment and application of the model. This poses more problems for the encrypted traffic identification task.

- Q1: How to design a mechanism to help the model automatically extract long-term stable features for Tor traffic flows encrypted at multiple layers while reducing the required computational resources?

- Q2: How to maintain the high robustness of Tor traffic flow classification when facing disorders and packet losses in the network?
- Q3: How to keep the high identification accuracy under dynamic changes in network traffic data?

Considering the above problems, we convert the problem of encrypted traffic identification into an image recognition problem in computer vision and propose our novel framework SRViT, as illustrated in Fig. 3. Our framework mainly consists of a preprocessing module, an embedding layer with a multi-size patch mechanism, an encoding layer containing a Bias Unit, and an output layer. To address the first problem, we design an overlapping multi-size patch mechanism in the embedding layer to learn the features of encrypted traffic flows more comprehensively. By analyzing patches of multiple sizes, the multi-size patch mechanism ensures that both features at different granularity levels, including fine details and overarching patterns, are captured simultaneously, thus providing richer feature sets for the model to improve classification accuracy and robustness. In addition, to solve the problem of Tor's random routing or packet disorder when the network condition is poor, we propose a novel self-attention operation in the encoding layer to dynamically adjust the attention score and learn the relative position relationship in each encrypted traffic flow. Finally, we design an incremental update mechanism to quickly adapt our framework to the dynamic network environment by updating only a small proportion of parameters when the framework suffers from a large degradation in classification accuracy.

C. Preprocessing Module

This module is mainly responsible for transforming the real-time traffic in the format of pcap into the input data that meets requirements. We intercept the information from the raw flow data to form a two-dimensional matrix, which is processed as an image and categorized by the vision transformer algorithm. Different layers of encryption may affect the traffic differently, creating complex multi-dimensional challenges for feature extraction and analysis. The reason for representing the flow as a two-dimensional matrix rather than one-dimensional is to utilize the second dimension to depict the multilevel information of the Tor flow [30]. Encrypted network traffic often exhibits temporal patterns. ViT can capture these patterns and correlate them with spatial patterns in images.

For encrypted traffic data, a lot of useful information is lost if only the packet size sequence is utilized. Especially for the Tor encrypted traffic, the characteristic of fixed-size cells impairs the effectiveness of the methods that use only packet sizes. During TLS communication, the client and server exchange handshake plaintext data to negotiate the parameters used in subsequent encrypted connections, such as version, cipher suite, extensions, and certificates. Previous research has shown that this information is very effective in encrypted traffic recognition [51]. We convert the bytes of the encrypted network flows into images, with one byte corresponding to one pixel of the image. Although encryption makes the payload more randomized, most of the original information in the network traffic can be retained since the image features

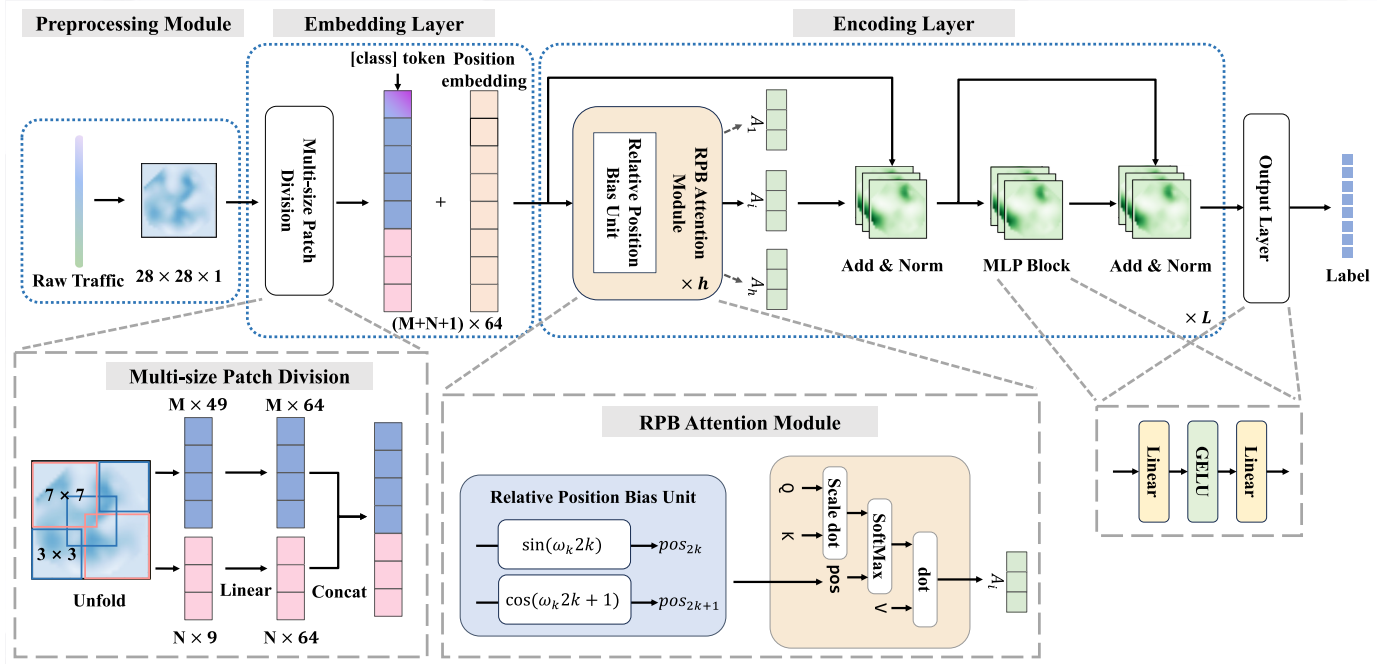


Fig. 3. Overview of our method SRViT.

are extracted from the raw data of the encrypted traffic. As a result, the image features have received widespread attention and utilization in encrypted traffic identification research.

However, encryption would introduce noise in the image converted by the traffic, so choosing an appropriate image size is particularly important in the feature extraction process of the image. If the size chosen is too small, the image will not be able to fully describe the information of the encrypted traffic and if the size chosen is too large, the information redundancy will affect the accuracy and efficiency of the identification model. In order to keep the information of the encrypted traffic completely and also ensure high accuracy and efficiency, we select the first 784 bytes of the original encrypted traffic to be converted into a grayscale image for identification after sufficient experiments [52].

We recognize packets with the same IP address, port number, and protocol type as the same flow and only keep the first n bytes of each flow. Then, we format the byte sequence into an image of size $H \times W \times C$ through a reshaping operation, where H , W , C denote the height, width, and number of channels of the image, respectively. In this paper, n is taken as 784, and the intercepted encrypted traffic flow is converted to a grayscale map with a size of $28 \times 28 \times 1$. Fig. 2 displays some visual results of preprocessed network flows belonging to different types. From the visualizations, we conclude that the different classes of traffic are obviously discriminated from each other and have a high consistency across each class, so it is reasonable to assume that good performance can be obtained with our approach. At the same time, the differences between the traffic with Tor encryption seem to be much smaller compared to the traffic without Tor encryption, which impairs the effectiveness of directly using computer vision algorithms such as CNN, ViT, etc.

D. Multi-Size Patch Division

Multiple encryption layers obscure traffic patterns, making it difficult to analyze or classify traffic based on traditional feature extraction methods. Vision Transformer (ViT) processes input images by dividing them into a spatially regular grid of equal-size patches without overlapping. It hinders the application of ViT in network traffic identification and may lead to some information loss in feature extraction, especially for encrypted Tor traffic. Thus we propose a novel patch dividing mechanism by multi-size division with overlapping to complement the extracted features of the flow.

Image features are discrete, while network traffic features are continuous. This makes it possible that the direct application of ViT may miss key features of the traffic, especially for Tor traffic flows encrypted at multiple layers, potentially leading to decreased recognition accuracy. To solve the above problem, we design a simple multi-size patch mechanism that can well capture the features of Tor traffic, characterized by multi-layer encryption, highly obfuscated payload, and random routing. Multi-layer encryption can impact traffic features at various scales. Smaller patches in the multi-size patch mechanism might capture fine-grained, localized details that larger patches might miss, while larger patches can capture broader, more global patterns.

As shown in Fig. 4, the input image $x \in R^{H \times W}$ is first split into overlapping 2D patches of size $P \times P$ with a total number of p . We use two layers with two different patch sizes P_1, P_2 along with two different stride lengths S_1, S_2 to extract different feature information from the input data. These features are represented as $X_0^0 = [x_{p1}^1; x_{p1}^2; \dots; x_{p1}^N]$, $X_0^1 = [x_{p2}^1; x_{p2}^2; \dots; x_{p2}^M]$. The numbers of patches generated by each of the two layers are denoted as $N = HW/S_1^2$, $M = HW/S_2^2$, and patches are represented as $x_{p1} \in R^{N \times P_1^2}$, $x_{p2} \in R^{M \times P_2^2}$. The

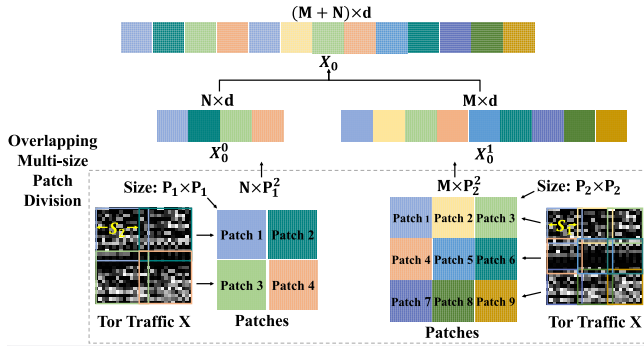


Fig. 4. The schematic illustration of multi-size patch division.

patches are then mapped to d -dimensional vectors by a linear layer as patch embeddings.

The patch size and stride length need to be set reasonably. If they are too large, they cannot capture the local information of the multi-layer encryption of Tor traffic effectively; if they are too small, they cannot learn the global information effectively, and this will lead to excessively long model training times. According to the characteristics of Tor traffic, we experimented with several sets of parameters, setting P_1, S_1, P_2, S_2 to 3, 2, 7, and 4 respectively. This configuration allows our model to capture the multi-layer encryption information of Tor traffic while maintaining short classification times.

Meanwhile, since traffic data transmission is highly related to sequential order, the position embeddings are added to the patch embeddings as the input to the traffic encoder to maintain the crucial position information. With fixed localization of packets, coupled with position embedding, our approach transcends location limitations to capture long-term stable dependencies in traffic. Additionally, similar to the *[class]* token in BERT, a learnable embedding also needs to be prepended to the sequence of embedded patches ($X_0^0 = x_{class}$). The equation for this step is shown in Eq. (1):

$$X_0 = [x_{class}; x_{p1}^1 E; \dots; x_{p1}^N E; x_{p2}^{N+1} E; \dots; x_{p2}^{N+M} E] + E_{pos}. \quad (1)$$

E. RPB Attention Module

Traffic data is highly related to order, especially since Tor-encrypted traffic is characterized by random routing, so the order between packets is particularly important. ViT only adds positional embeddings to the embedding layer to represent absolute positions, which is insufficient in many cases for Tor-encrypted traffic. In encrypted traffic, especially in the case of disorder and packet loss, the relative distances between different packet patches are obviously different. Thus position information with different relative distance values also needs to be assigned to different patches. Therefore, we propose to add a relative position bias to solve this problem.

However, if the relative position embedding is added directly to the embedding layer, after a series of matrix operations in self-attention, the relative position information is still lost and only the absolute position information remains. A natural idea is to add the relative position information back into the attention bias computation. Therefore, we introduce a relative position bias unit B_{RPB} in the multi-headed self-attention mechanism, in which patches can effectively interact

with each other based on the relevance of different heads $Concat(head_1, \dots, head_h)$, and each head is computed by the attention function:

$$MSA_B = softmax\left(\frac{Q \cdot K^T}{\sqrt{d}} + B_{RPB}\right) \cdot V, \quad (2)$$

$$Q = x_l W^Q, K = x_l W^K, V = x_l W^V,$$

where Q, K, V stand for Query, Key, and Value, respectively, and they are all matrices. $W^Q, W^K, W^V \in \mathbb{R}^{d \times d}$ are learnable parameters. d is the embedding dimension, which is used for normalization. Note that the bias unit is a trainable parameter. B_{RPB} needs to be calculated according to Equation Eq. (5) in the next section. It can be seen that B_{RPB} serves to add a value to each element of $Q \cdot K^T$, the essence of which is that it wants self-attention to be further biased. The relative position does not completely model the position information of each input, but the relative distance between the current position and the position that is being attended is considered when counting attention score, which can be used to adjust the scores of the attention between the different positions in the self-attention computation and make the model learn the relative position relationship.

Finally, the encoding layer consists of the RPB self-attention module and MLP block stacked in L layers. The mathematical format is shown in Eq. (3) and Eq. (4). To enhance the robustness and convergence speed of the model, layer normalization (LN) is applied before every block, and residual connections are applied after every block. The MLP Block includes the Fully Connected Layer and the GELU Activation Function.

$$X'_l = MSA_B(LN(x_{l-1})) + x_{l-1}, \quad l = 1 \dots L, \quad (3)$$

$$X_l = MLP(LN(x'_l)) + x'_l, \quad l = 1 \dots L. \quad (4)$$

F. RPB Unit

To address the challenge of traffic identification posed by the random routing characteristics of Tor traffic, we add relative position bias to the self-attention mechanism. Relative position bias makes it possible to discriminate the relative positions between different tokens by fine-tuning the self-attention arithmetic process. The aim is to make the position vectors of the same dimension at different positions contain the relative position information between them, which can be realized by the periodicity of the function. Therefore, trigonometric functions can be utilized to represent the relative position information, which is essentially the use of sine/cosine formulas with different frequencies for different dimensions thus generating high-dimensional position vectors for different locations. It is calculated by the following equation Eq. (5):

$$PE_{pos}^{(i)} = \begin{cases} \sin(\omega_k pos), & i = 2k \\ \cos(\omega_k pos), & i = 2k + 1 \end{cases}$$

$$\omega_k = \frac{1}{10000^{2k/d_{model}}}, \quad (5)$$

where the resulting PE is computed as a 2-dimensional matrix with pos corresponding to each input position and i representing the i th position of each position vector. d_{model} is

the vector dimension such that each position has an encoding that matches the d_{model} dimension. Note that in this paper the position vectors are summed with the results of the previous vectors and therefore the value is taken as the length of the sequence. The equation shows that different dimensions use sine/cosine functions with different frequencies to generate high-dimensional position vectors for different positions that express relative position information. The advantage of using trigonometric design in this way is that: firstly, the positional encoding at position i can be linearly represented by the positional encoding at position $i + k$, reflecting the relative positional relationship between the two positions. Secondly, the inner product of the positional encoding at positions i and $i + k$ decreases as the relative position increases, thus characterizing the relative distance of the positions. Finally, it ensures that the position encoding of each token is unique and regular, with small differences between neighboring positions and larger differences between positions that are farther away. Moreover, it ensures that the relative positions are bounded and located in a continuous space.

Algorithm 1 Soft Relative Vision Transformer (SRViT)

- 1: **Input:** Image X with dimensions $H \times W \times C$
 - 2: **Output:** Class predictions Y
 - 3: **Initialization:**
 - 4: Embedding dimension d , Number of attention heads h , Number of transformer layers L , Patch size p_1, p_2 .
 - 5: **Multi-size Patch Division:**
 - 6: Divide the image into overlapping patches, linearly project patch vectors to d -dimensional space and add positional encoding and class token as section IV-D.
 - 7: $X_0 = [x_{class}; x_{p1}^1 E; \dots; x_{p1}^N E; x_{p2}^{N+1} E; \dots; x_{p2}^{N+M} E] + E_{pos}$
 - 8: **RPB Attention Module:**
 - 9: **RPB Unit:**
 - 10: $PE^{(2k)} = \sin\left(\frac{pos}{10000^{2k/d}}\right)$
 - 11: $PE^{(2k+1)} = \cos\left(\frac{pos}{10000^{2k/d}}\right)$
 - 12: **for** $l = 1$ to L **do**
 - 13: **for** $i = 1$ to h **do**
 - 14: $MSA_{Bi} = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d}} + B_{RPB}\right) \cdot V$
 - 15: **end for**
 - 16: $X'_i = LN(X_{i-1})X_{i-1}$
 - 17: $X_i = MLP(LN(X'_i)) + X'_i$
 - 18: **end for**
 - 19: **Output Layer:**
 - 20: **return** $Y = LN(X_L^0)$
-

G. Output Layer

Since the task of this paper is to classify the encrypted traffic, the role of this module is to extract the corresponding $[class]$ token to output classification results. The shape of the output tensor after passing through the Transformer Encoder remains the same as the input tensor. The output layer in Vision Transformer is composed of a fully connected layer and GELU activation and a fully connected layer. The detailed calculated formulation is as follows (Eq. (6)),

$$Y = LN(X_L^0). \quad (6)$$

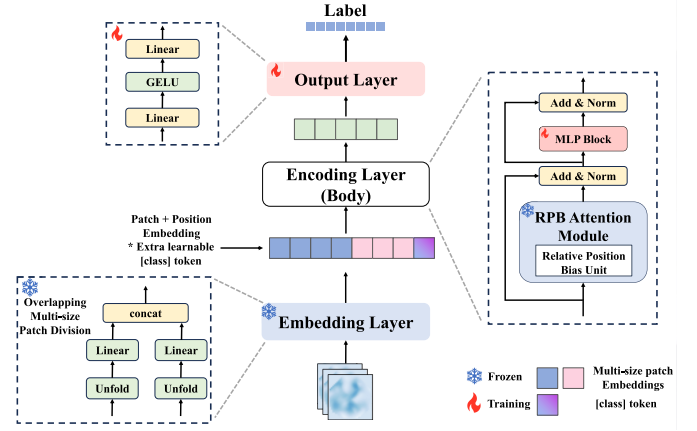


Fig. 5. The schematic diagram of the Incremental Update Mechanism.

H. The Incremental Update

Since Tor traffic is characterized by uncertain traffic patterns and the nature of Internet traffic flow is dynamic over time, therefore we design the process of an incremental update to adapt to the changing characteristics. When a significant drop in the accuracy of traffic recognition is monitored, our model determines that there is an obvious change in flow representations, indicating the current model parameters are no longer applicable. However, retraining the model is expensive, the incremental update mechanism updates the specific parameters of the model to achieve accurate online traffic classification without retraining the whole model.

Layers closer to the bottom encode generic reusable features, whereas encoding layers closer to the top may focus additionally on task-specific context and semantics, so the top layers capture more abstract features and encode more specialized features that are better adapted to the specific task. Updating these specialized features is more useful. Specifically speaking, we freeze most of the parameters of the model and update only the parameters of the MLP layer for each Transformer encoder layer. As illustrated in Fig. 5, the other parts of the model (depicted as blue blocks in Fig. 5) are loaded with weights from the original model, and the parameters are kept frozen. The MLP layer parameters (depicted as red blocks in Fig. 5) are updated on a new task-specific data domain containing task-specific losses.

In this paper, we detect the output probability distribution when the model classifies the current traffic, which usually reflects the model's confidence that the traffic is classified into the current category. We set the threshold of the maximum output probability to 0.9, which means that the incremental update mechanism is triggered when the average of the model's maximum classification probability for the current batch of data is detected to be below 0.9. A decrease in the output probability when the model classifies may suggest a decrease in the model's confidence in classifying the current traffic data, so we train the original model with incremental updates using newly arrived traffic.

V. EXPERIMENTS

In this section, we perform four encrypted flow classification tasks to demonstrate the effectiveness of the SRViT model

TABLE II
DATASETS

No.	Dataset	flow number		Size	label
		Train	Test		
Dataset 1	USTC-TFC	245437	27271	3.71GB	20
Dataset 2	ISCX-VPN	41822	4647	28GB	12
Dataset 3	ISCX-Tor	66454	27400	22GB	14
Dataset 4	CIC-IoT2022	4813	535	1.47GB	4
Dataset 5	VNAT	17163	1907	36GB	10
Dataset 6	Gotham2025	28438	3159	23.8GB	6

in solving different encryption scenarios and data distribution imbalance problems. In addition, we perform ablation experiments to verify the effects of our key designs in SRViT. We also compare our model with the state-of-the-art models in terms of robustness, online classification performance and execution effectiveness.

A. Experiment Setup

1) *Datasets and Classification Tasks*: To assess the effectiveness and generalization of SRViT, we conducted experiments on encrypted traffic classification tasks using five different types of public datasets. The corresponding datasets are presented in Table II.

Dataset 1: The USTC-TFC dataset [53] comprises encrypted traffic samples consisting of both malicious software and benign applications. It is utilized for the task of encrypted malicious traffic classification, consisting of a total of 20 categories, including 10 benign and 10 malignant traffic types. The size of the USTC-TFC dataset is 3.71 GB.

Dataset 2: The ISCX-VPN dataset [54] is composed of 6 communication applications in both VPN and non-VPN scenarios captured by the Canadian Institute for Cybersecurity. It is designated to classify encrypted traffic that uses Virtual Private Networks (VPNs) for network communication. The dataset contains 14 types of raw traffic data, consisting of 7 types of non-VPN and VPN traffic respectively.

Dataset 3: The ISCX-Tor dataset [55] is a labeled dataset of Tor traffic published by the University of New Brunswick (UNB). It aims to classify encrypted traffic that uses an Onion Router (Tor) to enhance communication privacy. The dataset contains 16 traffic classes, 8 categories of Tor traffic, and 8 categories of non-Tor traffic. This traffic obfuscates the communication between senders and receivers through a distributed routing network, further obscuring the behavior of the traffic, and thus enhancing the difficulty of traffic classification.

Dataset 4: The CIC-IoT2022 dataset [56] is a multi-purpose academic IoT dataset for profiling, behavioral analysis, and vulnerability testing of different IoT devices. We selected realistic traffic flows captured in interaction experiments, including 4 different interaction methods for 28 devices. Categorizing by device type generates the dataset containing 4 different device types.

Dataset 5: The VNAT dataset [57] is a VPN/non-VPN network application traffic dataset provided by MIT Lincoln Laboratory, which primarily contains labeled PCAP data files covering encrypted and non-encrypted traffic data for

10 applications across 5 traffic categories. The dataset is divided into 10 categories, containing 5 categories of VPN and 5 categories of non-VPN traffic data.

Dataset 6: The Gotham2025 dataset [58] is a large-scale IoT network traffic dataset based on the Gotham testing platform. It contains traffic generated by 78 simulated IoT devices, covering protocols such as MQTT, CoAP, and RTSP. The dataset includes 1 category of benign and five distinct types of malicious traffic.

2) *Data Pre-Processing*: In this paper, the data preprocessing is carried out according to Section IV-C to convert flows into grayscale images. The datasets are split by the five-fold cross-validation to guarantee the credibility of the experimental results.

3) *Evaluation Metrics*: Generally four evaluation metrics: accuracy, precision, recall, and F1 are used to evaluate and compare the performance of the models. Accuracy is used to evaluate the overall performance of the classifiers. In the case of class imbalance, precision and recall need to be considered, while the F1 score balances the relationship between precision and recall, therefore making it more robust compared to accuracy. TP represents the number of correctly classified positive samples, TN represents the number of correctly classified negative samples, FP represents the number of instances incorrectly classified positive samples, and FN represents the number of instances incorrectly classified negative samples. The formulas are as follows:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}, \quad (7)$$

$$Precision = \frac{TP}{TP + FP}, \quad (8)$$

$$Recall = \frac{TP}{TP + FN}, \quad (9)$$

$$F1 = 2 \cdot \frac{Precision \times Recall}{Precision + Recall}. \quad (10)$$

4) *Implementation Details*: For the training phase, we use the Adam optimizer for 100 epochs with the base learning rate set to 1×10^{-3} and a batch size of 32. The batch size is 64 for testing. We implemented the proposed method using PyTorch 2.1.1 and trained it on the GPU environment of NVIDIA GeForce GTX3090.

In time performance experiments, we compare and analyze the time performance of the ET-BERT algorithm and our algorithm respectively in a GPU environment, which is an NVIDIA GeForce GTX3090.

B. 7 State-of-the-Art Models

We compare SRViT with 7 various state-of-the-art models, including:

- ET-BERT [28] treats traffic representation extraction as an NLP task for pre-training and then fine-tunes the classifier with limited labeled data. It's important to note that the ET-BERT model requires pre-training. In reproducing the experiments, we directly downloaded the pre-training files provided by the original authors and trained only in the fine-tuning phase.

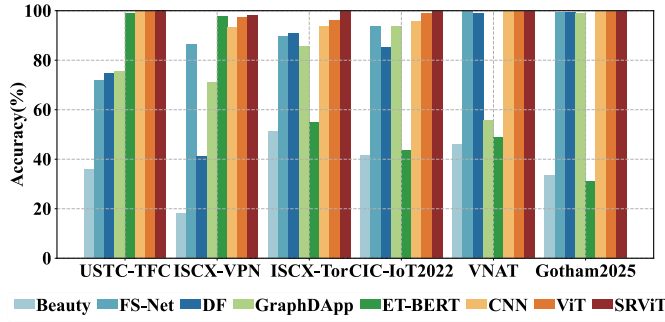


Fig. 6. The results of accuracy comparison.

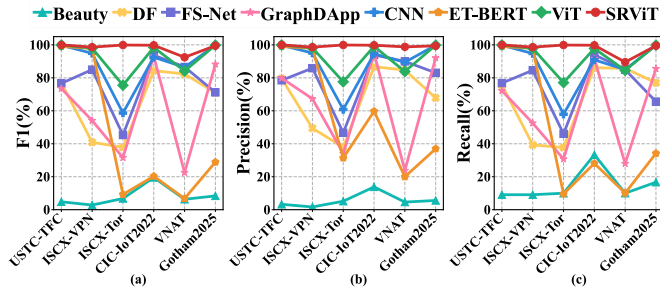


Fig. 7. Performance comparison on 6 datasets.

- FS-Net [20], Deep Fingerprinting (DF) [24], GraphDApp [45], Beauty [44], and 2D-CNN [52] are DL-based traffic analysis methods, which is training by the supervised way using raw packet information.

C. Accuracy Comparison

The experimental results of accuracy performance are shown in Fig. 6 and Fig. 7.

Encrypted malicious traffic: The results for USTC-TFC indicate that our model outperforms all other methods on four metrics, with our model achieving an accuracy of 99.99%. The malicious traffic in this dataset contains application-layer unencrypted data, which makes it easier for other models to utilize such plaintexts for classification.

VPN encrypted traffic classification: The VPN encrypted traffic classification contains two datasets. On the ISCX-VPN dataset, our method outperforms other models for all metrics, and we achieve the best performance on F1, which can reach 98.61%. In addition, this dataset poses the challenge of data imbalance to classification algorithms. The experimental results suggest that our model exhibits better performances in identifying obfuscated traffic even in the presence of data imbalance. Meanwhile, the accuracy of our algorithm can still reach 99.74% on the VNAT dataset. As can be seen from the figure, our algorithm has the best results on all metrics.

Tor encrypted traffic classification: The results on ISCX-Tor show that our algorithm results in a 45.17% accuracy improvement over the ET-BERT model, which holds the best performance among the models we compared. Our method can achieve 99.96% accuracy, 99.89% recall, 99.91% precision, 99.90% F1. Moreover, on this dataset, our method achieves an average improvement of 22.31% accuracy, 63.89% precision, 67.91% recall, and 68.33% F1 over the previous algorithms,

respectively. Furthermore, we observe that DL-based methods perform poorly, whether using sequence features or payloads as inputs. The reason is more likely that the initial traffic in Tor is not only encrypted with multiple layers, but also carried out antagonistic obfuscation, making it difficult to analyze the payload directly, and the sequence features hardly work on limited captured packets. Meanwhile, Tor traffic is characterized by variable packet lengths the use of intercepting raw bytes does not help discriminate each type of traffic encrypted with Tor and cannot differentiate the traffic information, while our multi-size patch mechanism can solve this problem by extracting features in multiple layers.

IoT traffic classification: As shown in the results, our method outperforms all previous models in all metrics, including ET-BERT. ET-BERT achieves similarly excellent results to our model on the first three datasets. This is because ET-BERT needs to be pre-trained with a large amount of data, and the traffic of the first three datasets is included in the pre-training of the ET-BERT model. However, the traffic of this dataset is not included in the pre-training, so ET-BERT does not have the related knowledge of the IoT traffic. Our model does not need the pre-training process and still obtains 99.63% accuracy on the CIC-IoT2022 dataset and 99.62% on the Gotham2025 dataset.

D. Robustness Study

In real deployments, the data acquisition end often has the situation of out-of-order and packet loss, which will affect the identification method of encrypted traffic negatively. In this experiment, we randomly modified some of the traffic packets to explore the robustness of the algorithms in harsh and high-noise conditions. We conducted two sets of experiments: (1) packet disordering: randomly selected part of the packets to disrupt the order, and (2) packet loss: randomly selected part of the packets to be discarded. We set the ratio range of noise labeling from 0.2 to 0.8.

Both FS-Net and CNN are classical DL-based traffic classification algorithms, CNN leverages the spatial features of the traffic while FS-Net uses RNN which can focus on the temporal features of the traffic, therefore we select these two methods as the representatives and compare our approach with them. We demonstrate the results obtained by our method, the CNN model, and FS-Net on the Tor dataset in the presence of noise, and similar results are obtained on other datasets.

In Fig. 8, the first and last four plots show the experimental results of packet disordering and packet loss, respectively. The experimental results indicate that each model shows a different degree of degradation in the robustness experiments. Compared with the previous traffic classification model, our model is more fault-tolerant with an average improvement of 13.68% in accuracy, 54.83% in precision, 56.91% in recall, and 55.83% in F1 at a noise ratio of 0.8 in the case of packet disordering. Even when the noise rate reaches 0.8, SRViT still has strong classification capability.

We can observe that the performance of our model decreases the least in the case of equal noise. It is because at first, the mechanism of multiple layers of patches with different sizes is used to minimize the impact by dividing the packets into

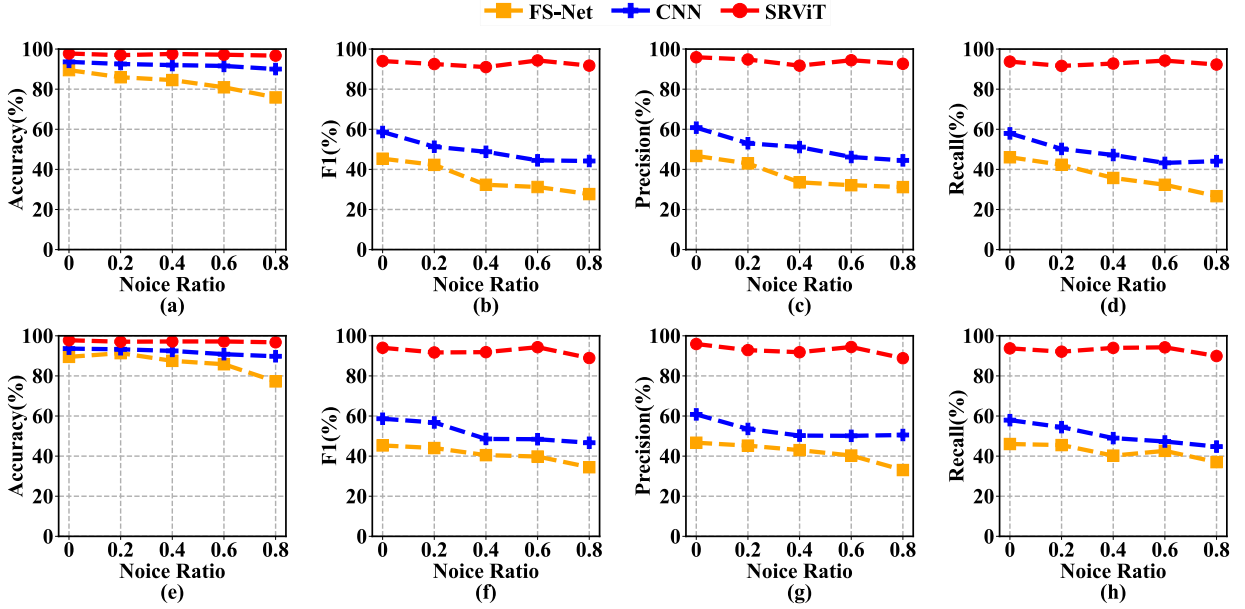


Fig. 8. Performances on robustness.

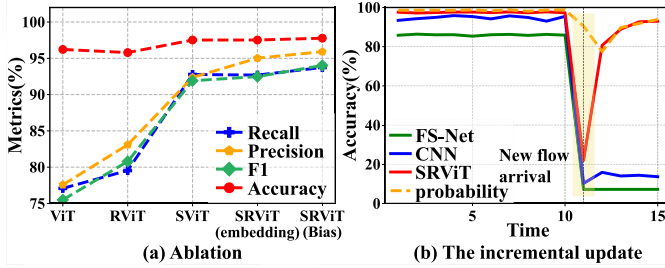


Fig. 9. Performance of ablation experiments and incremental update.

the same patch even when they are disordered. Secondly, we introduce the relative position bias into the model, making the classification performance more stable, while the absolute position instead brings about adverse impacts in case of disorder or packet loss.

E. Ablation Study

To verify the contribution of the multi-size patch mechanism and relative position bias in solving the problems of multi-layer encryption, highly obfuscated payloads, random routing, and traffic pattern uncertainty for Tor-encrypted traffic, we show the results of the ablation experiments in Fig. 9(a). ViT denotes the algorithm that utilizes the Vision Transformer model directly. SViT refers to the algorithm that incorporates multi-size overlapping patches on top of the Vision Transformer model. Then, RViT denotes the algorithm that introduces relative position bias to the Vision Transformer model. Furthermore, SRViT (embedding) combines multi-size overlapping patches and directly includes a relative position embedding in the Vision Transformer model. Lastly, SRViT (Bias) adds multi-size overlapping patches and includes a relative position bias in the Vision Transformer model.

We observe that the highest performance of the model is achieved after combining the two mechanisms introduced. First, the accuracy of our model increases to 97.52% after

the introduction of the multi-size patch mechanism. Second, from the results, we can see that the accuracy of SRViT (embedding) is the same as that of SViT, which can also prove that by adding relative position embedding directly, the relative position information is still lost in the calculation of attention scores, and thus the performance is not significantly improved. However, the performance of the classifier improves when relative position bias is introduced. In conclusion, the two mechanisms we introduce can extract deeper long-term stable features of encrypted traffic as well as inter-packet position information to improve the classification performance of the model.

F. Online Classification Performance

In this section, the results of the incremental updating experiments are shown to verify the effectiveness of our online performances. We first train the SRViT model using the VPN dataset with 100 epochs and test it, and then test it using the other datasets to simulate the long-term change of traffic. Through the proposed mechanism of incremental update, our model could adapt to these dynamic changes and keep a stable performance, compared with other methods. Fig. 9(b) shows the results of the last 10 tests and the results of 5 tests after using the incremental update. During online classification, we test the output probability of the model. In our experiments, we use labeled datasets and show the accuracy as a comparison. As can be seen from Fig. 9(b), the output maximum probability of the model suffers a large drop when traffic arrives from a new dataset, and the accuracy rate drops dramatically at that time. Therefore, we may determine whether the model needs to be an incremental update by detecting the maximum output probability. After this, the incremental update can quickly recover the model's classification accuracy, and the output probability also gradually recovers in the same trend as the accuracy. Similar results were found on other datasets. Furthermore, to validate the robustness of the

TABLE III

NUMBER OF PARAMETERS & FLOPS OF THE TWO METHODS

	Total	Trainable	FLOPs(G)
SRViT	1.16×10^6	1.16×10^6	3.15
ET-BERT	1.32×10^8	1.32×10^8	347.91

TABLE IV

THE GPU ENERGY CONSUMPTION

	SRViT	ET-BERT	Percentage reduction
GPU Fan	50%	64%	21.88%
GPU Temp (C)	56	65	13.85%
Power Usage/Cap (W)	236/350	340/350	30.59%
Memory-Usage (MiB)	1438/24576	7136/24576	79.85%
GPU utilization rate	70%	99%	29.29%

confidence-based update mechanism, we conducted experiments following the robustness test setup described in Section V-C. Results demonstrate that even under packet loss or out-of-order conditions, our method maintains high confidence above the trigger threshold.

G. Time Consumption Comparison

The goal of this paper is to propose a transformer-based traffic classification method to maintain the model classification performance while substantially reducing the required computational resources. Therefore, we compare the time performance energy consumption with the transformer-based model ET-BERT. In this section, we display the results of the time performance on the ISCX-VPN dataset, and the same results are obtained on the other datasets.

1) *The Number of Parameters*: FLOPs is the number of floating point operations, a measure of the computational complexity of a model/algorithm. The larger the FLOPs, the higher the computational complexity of the model. Therefore, we compared the number of training parameters and FLOPs of our model and ET-BERT. Our model has a total parameter count of 1.16×10^6 , with all of them being trainable, while the total number of parameters and the number of trainable parameters of the ET-BERT model is 1.32×10^8 . Moreover, our model has FLOPs of 3G, while ET-BERT has FLOPs of 348G, indicating that our model is more lightweight and convenient to deploy. Comparison results are shown in the Table III.

Although FLOPs is a commonly used evaluation metric for measuring algorithms today, FLOPs does not take into account MAC (Memory Access Cost) and parallelism, etc., so it may not fully represent the model speed and energy consumption. We also conducted the following experiments.

2) *The GPU Energy Consumption*: We compare the energy consumption of the two models deployed on the same GPU. The experimental results in the Table IV demonstrate that our results outperform ET-BERT in all metrics, where the average GPU power, SRViT, is 30.59% lower than ET-BERT, and the average GPU memory is 79.85% lower. This indicates that our algorithm can reduce the resource consumption more while maintaining a higher performance.

3) *Training Time*: The total training time of the model refers to the total time spent from feeding data into the model

TABLE V
TRAINING TIME ON GPU

	SRViT	ET-BERT	Percentage reduction
Total time(s)	5266.79	6081.29	13.39%
Time per Epoch(s)	406.02	2962.75	86.30%

TABLE VI
TESTING TIME ON GPU

	SRViT	ET-BERT	Percentage reduction
Total time(s)	33.88	85.61	60.43%
Time per flow(ms/flow)	7.29	14.27	48.91%

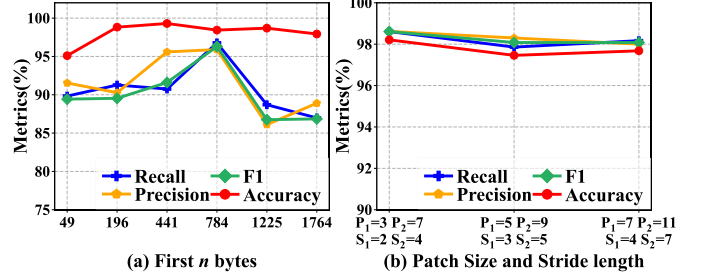


Fig. 10. Influence of number of bytes, patch size and stride length.

until after completing model convergence, and the time per epoch is the average of the time it takes for the model to train one epoch.

In the ET-BERT experiments, the total training time does not include the time spent in the pre-training phase, but only the time spent in the fine-tuning phase using the transformer model to classify the encrypted traffic. The experimental results are shown in Table V, which indicates that the training time of our model is much shorter than that of ET-BERT if pre-training is included. The last column in the table represents the percentage reduction in time.

4) *Testing Time*: The testing time means the time it takes to output the classification results of the encrypted flows on the test set. We divide the total time by the number of flows contained in the test set to get the time to classify each flow. As can be seen from Table VI, the classification time of ET-BERT is approximately twice as slow as that of our method.

H. Influence of First n Bytes

To select the appropriate number of the first n bytes, we have chosen different n for the experiments. Fig. 10(a) shows the experimental results on the Tor dataset.

We randomly selected half of the traffic in the Tor encrypted traffic dataset for our experiments. From the Fig. 10(a), we can observe that the accuracy can achieve 98.44% when n is taken 784. Moreover, optimal results are achieved on the other three metrics, with average improvements of 5.42% on recall, 7.26% on precision, and 7.47% on F1 compared to taking other byte numbers. According to the experimental results, with n set at 784, each metric reaches or closely approaches its optimal value, and a balance can be achieved among the metrics. We achieve similar comparison results on other

3 datasets. Therefore, we can confirm that setting n to 784 is suitable to reach optimal values of metrics while maintaining high efficiency.

I. Influence of Patch Size and Stride Length

In order to select the appropriate patch size and stride length, we perform comparative experiments by selecting different groups of patch sizes and stride lengths, respectively. The stride length varies with the patch size. Fig. 10(b) shows the experimental results on the VPN dataset.

The Fig. 10(b) illustrates that when P_1, S_1, P_2, S_2 take the values of 3, 2, 7, and 4, better results can be achieved for all the metrics. Among them, the accuracy can reach 98.21%. We obtained similar results on other datasets. This is because the convolution can extract more local features and improve the identification accuracy when the value of patch size is taken smaller. Therefore, we confirm that setting P_1, S_1, P_2, S_2 to 3, 2, 7, and 4 respectively is appropriate to achieve the optimal values for each metric.

VI. DISCUSSION

In this section, we discuss the limitations of this work, as well as further research efforts in this area. We convert the problem of encrypted traffic flow recognition into an image recognition problem in computer vision and propose a novel end-to-end recognition model SRViT. We propose a novel multi-size patch division mechanism, a relative position bias and an incremental update mechanism. However, there are still some issues to be further investigated in this work. Firstly, since different classes of encrypted traffic have different types of packets, the more appropriate byte number, patch size, and stride length still need to be further investigated. One future work is to explore an adaptive or data-driven selection of these parameters. For example, different types of traffic or different applications might benefit from different image sizes (larger image for flows that carry more data, or smaller for flows with less data). Secondly, while SRViT has good robustness in various encrypted traffic scenarios, it relies on clean pre-training data. The attacker can intentionally add low-frequency mislabeled traffic data to carry out a poisoning attack against the model, and the decision boundaries of the model will be modified, thus spoofing the model used for a specific classification task and leading to a degradation of the accuracy. In addition, newly arriving traffic data in the incremental update mechanism may also cause the model to have odd shifts or catastrophic forgetting, thus we need to focus further on more robust training techniques.

VII. CONCLUSION

We introduce ViT into the encrypted traffic classification task and propose SRViT, a novel end-to-end identification model that transforms traffic flows into image patches and dynamically learns local/global dependencies between packets. We propose a novel multi-size patch division mechanism to further comprehensively learn the hidden features and resolve the multi-layer encryption, highly obfuscated payloads, random routing, and traffic pattern uncertainty problems. In

addition, we propose a self-attention operation with a novel relative position bias, which can dynamically adjust the attention scores and learn the relative position relationship. After that, we design a novel incremental update to adapt quickly to the dynamically changed traffic. At last, the comprehensive experiments on 5 real-world encrypted traffic datasets are carried out. The experimental results indicate that our SRViT outperforms the state-of-the-art methods with an average accuracy improvement of 24.62% while keeping higher robustness and execution efficiency. Furthermore, the effectiveness of our proposed key designs for classifying encrypted traffic is demonstrated in ablation experiments.

However, there are still some issues to be further investigated in this work, such as the deployment of SRViT and security performance, adaptive parameter selection, and more robust training techniques. In future work, we will further investigate the ability of SRViT to resist attacks, as well as few-shot learning or unsupervised learning methods.

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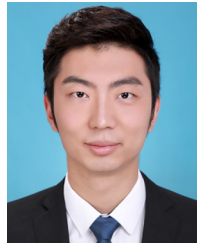
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